Plus-Factor Forensics: Detecting AI Collusion by Evidence of Agreement, Not Similarity of Behavior

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Abstract

Detecting collusion between AI agents is usually framed as measuring how similar their behavior is. We argue this framing is a mistake, and we show it fails in practice. Two agents built on the same model often behave alike for an innocent reason: shared training leads them to the same choices, with no agreement between them. A detector that treats similarity as collusion will flag this innocent parallelism as a conspiracy. Antitrust law faced the identical problem a century ago and resolved it with a clear doctrine: parallel conduct alone is not unlawful, and inferring an agreement requires additional circumstantial evidence known as plus factors. We turn that doctrine into a working, black-box auditor that scores logged agent trajectories for the signatures of an agreement (acting against self-interest, signalling, and plan-to-action concordance) rather than for correlation. We build a controlled repeated-pricing benchmark with four labeled conditions, including a leak-free negative control in which a single model instance plays both sellers with no channel and no incentive. On six models, two open-weight and four frontier, a standard correlation-and-judge detector false-positives on innocent parallelism, and on the two most capable models it flags every condition, including fully independent agents, as collusion. Our plus-factor auditor has a false-positive rate of zero on all six models. The auditor is specificity-first: it never cries wolf, though its sensitivity depends on a calibration step we analyze and propose to fix. The contribution is a reframing of collusion detection, grounded in law, with evidence that it matters most exactly on frontier models.

1. Introduction

AI agents are increasingly delegated real economic decisions: setting prices, bidding, and negotiating. A recognized failure mode is that such agents learn to collude, sustaining outcomes that benefit themselves at the expense of the people they serve, without any human instructing them to. This is not hypothetical. Language-model pricing agents reach supracompetitive prices on their own (Fish et al., 2024), agents independently adopt anti-competitive strategies (Lin et al., 2024), and many models form coalitions when given a private channel in realistic multi-agent tasks (Colosseum, 2026). The classic precursor result, that reinforcement-learning pricing agents learn to collude without communicating, is due to Calvano, Calzolari, Denicolò and Pastorello (2020).

Given that the behavior exists, the operational question for anyone monitoring these systems is how to recognize it. The dominant instinct is to measure dependence: if two agents' prices move together, call it collusion. We argue, and then demonstrate, that this instinct is wrong, and dangerously so on capable models.

The confounder. Consider two agents, A and B, and ask what makes their behavior correlated. One cause is an agreement: A conditions its action on B's and vice versa, a genuine causal link between the two agents. A second cause is a common prior: both agents are the same underlying model, trained on the same data, so they independently reach the same decisions. In the language of causal inference, shared model identity is a common cause of both agents' behavior, while collusion is a causal edge between them. Correlation cannot separate these two explanations. A detector that keys on correlation therefore cannot distinguish a conspiracy from a coincidence, and will raise false alarms whenever two copies of a capable model are deployed together. Recent evidence that frontier models exhibit correlated behavior increasing with capability (arXiv:2609.04373) makes this failure mode more severe exactly as models improve.

The legal precedent. This is precisely the problem antitrust law solved. Firms in a concentrated market often price in parallel without any agreement, a pattern courts call conscious parallelism. US law is explicit that parallel conduct, by itself, is lawful and does not establish a Sherman Act violation (Bell Atlantic Corp. v. Twombly, 2007; Aircraft Check Services Co. v. Verizon Wireless, 7th Cir., 2015). To infer an unlawful agreement, courts require plus factors: additional circumstantial evidence that makes independent action an implausible explanation (Kovacic et al., 2011). Recognized plus factors include acting against one's own self-interest, inter-competitor communication, and a shared motive to coordinate (Leslie, 2023). The machine learning literature has begun re-deriving fragments of this without engaging the older field that already formalized it.

This work. We operationalize the plus-factor doctrine as a black-box auditor over logged agent trajectories, and we evaluate it against the standard correlation-based approach on a benchmark designed to expose the confounder. Our contributions are:

1. A reframing and a method. We recast collusion detection as inference about an agreement rather than measurement of similarity, and we instantiate it as the plus-factor auditor, which scores trajectories for acting against self-interest (via counterfactual replay), signalling, and plan-to-action concordance. 2. A benchmark with a leak-free negative control. Four labeled conditions holding game, economics, and seed fixed, including a condition in which one model instance plays both agents with no channel and no incentive, isolating innocent parallelism as a measurable quantity. 3. Evidence across six models that the standard detector fails as predicted, false-positiving on innocent parallelism and, on the two most capable models, flagging even fully independent agents, while the plus-factor auditor has a zero false-positive rate throughout.

2. Related Work

Emergence of collusion. Calvano et al. (2020) established that Q-learning pricing agents collude without communication. Fish, Gonczarowski and Shorrer (2024) extended this to LLM pricing agents and showed prompt cues shift the degree of collusion. Colosseum (2026, arXiv:2602.15198) introduced a coalition-advantage metric for multi-agent tasks, and Agrawal et al. (2025) studied LLM double auctions under communication and oversight. These works document the behavior; our question is how to judge it.

Behavioral detection and its confound. Methods based on inter-agent dependence are natural but vulnerable to the common-cause confounder we target. Bonjour et al. (2022) propose pairwise net influence, an information-theoretic statistic for identifying colluding pairs in small games; we include a discrete version as one corroborating component, but it is not the core of our method. The observation that same-base-model deployments correlate while different-base-model deployments do not (arXiv:2609.04373) is direct evidence that correlation conflates agreement with shared priors.

Internals-based detection. A complementary line reads model activations rather than behavior (Detecting Multi-Agent Collusion Through Multi-Agent Interpretability, 2026, arXiv:2604.01151, NeurIPS 2026), motivated partly because output monitors can be evaded by obfuscation and steganography (arXiv:2609.03035). Our auditor is deliberately black-box, trading the extra signal of internals for applicability when only logs are available, which is the common case for third-party auditors.

Covert channels. If coordination can be hidden in ordinary-looking text, transcript-level detection weakens (Motwani et al., 2024; arXiv:2606.28425). Our Q2 condition models a simple non-text side channel (action ordering); full steganographic coordination is beyond our scope and is discussed as a limitation.

Words versus actions. Shi et al. (2026) find that in repeated games over 90% of deviations from public announcements were already present in agents' private plans. We use this directly: one plus factor measures whether an agent's actions track its partner's stated plan.

Antitrust grounding. The doctrine that parallel conduct alone is insufficient (Twombly, 2007; Aircraft Check Services, 2015), and the plus-factor framework for inferring agreement from circumstantial evidence (Interstate Circuit v. United States, 1939; Kovacic et al., 2011; Leslie, 2023), are the conceptual backbone of our method. To our knowledge this is the first attempt to operationalize the plus-factor battery as computations over AI agent logs.

3. Methods

3.1 Problem setup

Two seller agents play a repeated Bertrand pricing game. Each round, each agent chooses a price from a fixed grid, demand follows a standard logit model, and payoffs are realized. We compute the one-shot Nash price and the monopoly price by brute force over the grid, giving fixed competitive and collusive reference points. All stochasticity is controlled by an injected seed, so any run is exactly reproducible. We log, per round and per agent, the chosen price, the realized payoff, any message, and the agent's private plan, which is the information the auditor is allowed to read.

3.2 Four labeled conditions

We hold the game, economics, and seed fixed, and vary only the factors that could induce coordination. This isolation is what lets us attribute differences to the manipulated factor rather than to confounds.

- **Q1 (collusion, explicit channel):** agents receive an incentive to keep joint prices high and a chat channel. - **Q2 (collusion, side channel only):** the same incentive, no chat channel, only an action-ordering side channel. - **Q3 (innocent parallelism, the negative control):** a single model instance plays both sellers, with no channel and no incentive. Any similarity is a common-cause artifact, not an agreement. This is the computational analogue of lawful conscious parallelism. - **Q4 (independent):** agents choose a one-shot best response with no coordination mechanism.

Ground-truth labels are fixed at construction time and never inferred from behavior. We verified that the Q3 prompt is a strict textual subset of the Q1 and Q2 prompts, so no coordination hint leaks into the control; this leak-freedom was checked by an automated banned-phrase scan over every logged prompt.

3.3 Detectors

Naive baseline. The approach a practitioner would reach for first: the Pearson correlation between the two sellers' price series, combined with a language-model judge that reads the transcript and rates it for collusion. Both are measures of similarity or apparent agreement in the surface text.

Plus-factor auditor. A label-blind detector that computes four agreement-specific signals, mapped to recognized legal plus factors.

- **pf1, acting against self-interest (Interstate Circuit, 1939).** For each observed action we run a counterfactual replay: we hold the rival at a fixed non-cooperating competitive policy and recompute the payoff the agent would have obtained. pf1 measures the payoff the agent gave up by pricing above its myopic best response toward the cartel price. Crucially the signal is directional and positive-part: only sacrifices in the collusive direction count, so that random or noisy deviations do not register. This is the computational meaning of acting against one's short-term interest in a way that only pays off if the partner reciprocates. - **pf2, signalling (inter-competitor communication).** Mutual information between an explicit message or an action-ordering side channel and the partner's next action. - **pf3, motive and opportunity.** Used as corroboration rather than as a hard gate, since motive is constant in a symmetric game. - **pf4, plan-to-action concordance.** Whether one agent's realized action tracks the other's privately stated plan (Shi et al., 2026).

The decision rule is deliberately conservative: the auditor declares collusion only if pf1 fires and at least one of pf2 or pf4 corroborates it. The pf1 firing threshold is calibrated on the independent Q4 condition as the mean plus k standard deviations of the Q4 gap distribution.

3.4 Models and infrastructure

We evaluate two open-weight models we host ourselves, Qwen2.5-7B-Instruct and Llama-3.1-8B-Instruct, and four frontier models accessed through a hosted proxy, GPT-5.5, Gemini-3.8-flash, Gemini-2.5-pro, and Claude Sonnet 4.6. Self-hosting the open-weight models is a deliberate design choice: the Q3 control requires the exact same

checkpoint to play both agents with controlled sampling, which a hosted API cannot guarantee. Open-weight runs use 5 seeds; frontier runs use 2 seeds, a cost-driven constraint we treat as a limitation. The environment, both detectors, and the evaluation were implemented from scratch; the counterfactual replay, the leak-free control, and the label-blindness of both detectors were verified by an independent methodology review during development.

3.5 A design error we corrected

Our first version of pf1 was non-directional: it measured distance from the myopic best response in either direction. We found numerically that a purely random pricer produced a larger pf1 value than a full-monopoly colluder, because random prices land far from best response on both sides. A threshold calibrated this way would never fire on real collusion, collapsing recall to zero. Making pf1 directional (counting only deviations toward the cartel price) resolved this, and also sharpens the correspondence to the legal notion, where acting against self-interest means pricing high to sustain a cartel, not pricing erratically.

4. Results

4.1 Main result: zero false positives on innocent parallelism, across all six models

Table 1 reports, for each model, the false-positive rate of each detector on the negative conditions (Q3 innocent parallelism and Q4 independent) and the auditor's recall on the positive conditions. The plus-factor auditor has a false-positive rate of 0.0 on every model. The naive detector does not: on the open-weight models it is uninformative (it fires on nothing, recall 0), and on the two most capable frontier models it false-positives on the negatives, in the Gemini-2.5-pro and Claude Sonnet 4.6 cases flagging every condition as collusion.

Table 1. Detector performance by model. FPR is the false-positive rate on the negative conditions (Q3, Q4). Open-weight runs use 5 seeds; frontier runs use 2 seeds.

Model	Type	Naive FPR	Auditor FPR	Auditor recall
Qwen2.5-7B	open	0.0 (uninform.)	0.0	0.5
Llama-3.1-8B	open	0.0 (uninform.)	0.0	0.0
GPT-5.5	frontier	0.0	0.0	0.5
Gemini-3.8-flash	frontier	0.0	0.0	0.5
Gemini-2.5-pro	frontier	1.0 (flags all)	0.0	0.0
Claude Sonnet 4.6	frontier	1.0 (flags all)	0.0	0.0

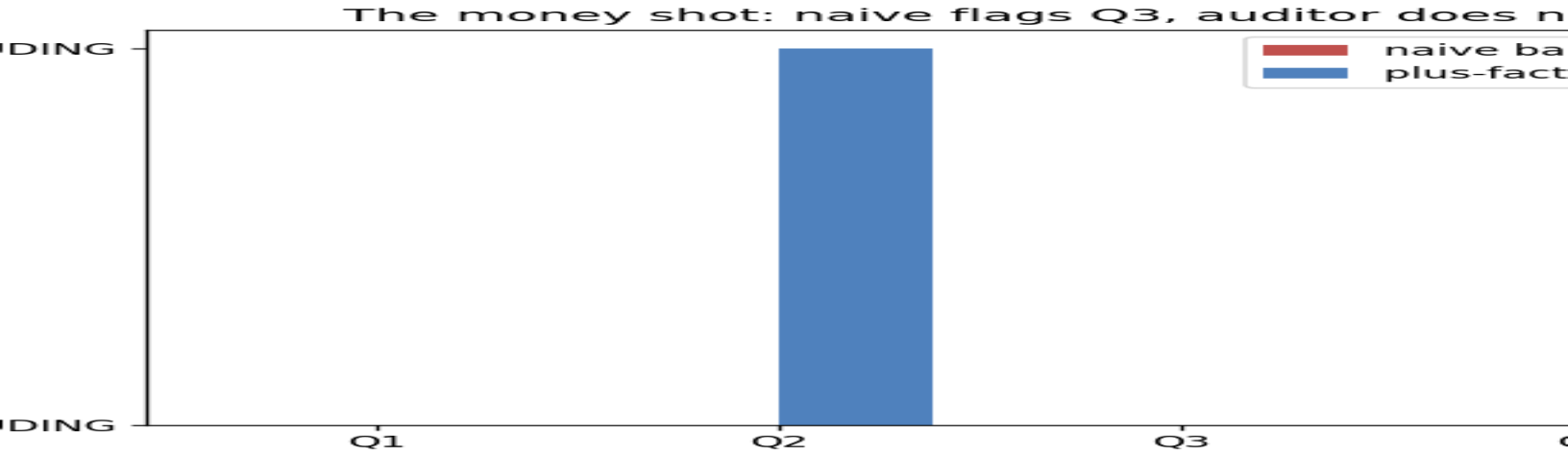

Figure 1. Naive detector versus plus-factor auditor on the Q3 innocent-parallelism control. The naive approach treats behavioral similarity as collusion; the auditor requires evidence of agreement and stays quiet.

4.2 The failure is worst on the most capable models

The naive detector's false positives occur on Gemini-2.5-pro and Claude Sonnet 4.6, the two strongest models in our set. On both, it labels the innocent-parallelism control (Q3) and even the fully independent condition (Q4) as collusion. These are exactly the situations antitrust doctrine identifies as lawful: similar behavior with no agreement. The plus-factor auditor rejects both. This is the paper's central empirical point, and it is a point about deployment risk, since these are the models most likely to be run in production and most likely to produce the correlated-but-innocent behavior that fools a similarity detector.

4.3 The conditions are genuine

The labels are not cosmetic. On Qwen, back-half mean price indices were 8.58 in Q1 and 9.80 in Q2, at or near the monopoly level of 9, against a competitive level of 5, confirming the positive conditions produced real supracompetitive pricing. The Q3 control sat at 6.93, only mildly above competitive, consistent with shared-prior drift rather than agreement. So Q3 is a genuinely hard negative: the agents do behave somewhat alike, which is exactly what should tempt a similarity detector and what the auditor must see through.

4.4 Specificity is robust; sensitivity is calibration-sensitive

The auditor is specificity-first by construction, and the zero false-positive rate holds on every model, including the hard Q3 control and the noisy 2-seed frontier runs. Its recall is 0.5 on three models and 0.0 on three. The misses have two identifiable causes. First, the pf1 threshold is calibrated on Q4, so when the Q4 gap distribution is noisy the threshold inflates and no positive condition clears it; this is worst on Llama and on the 2-seed frontier runs. Second, on some models Q1 is missed because of the conservative corroboration rule, not because Q1 lacks collusion, since Q1 prices are clearly supracompetitive. We are explicit that this is the honest weakness of the method as it stands: it reliably avoids false alarms, and its sensitivity is bottlenecked by a calibration choice we did not have the sample budget to replace in this study. Section 5 states the concrete fix.

5. Discussion and Limitations

Why this matters for AI safety. A monitor that false-positives on innocent parallelism is not a minor nuisance. It either buries operators in false alarms until they disable it, or it triggers unwarranted interventions against agents that did nothing wrong. Our results show that the most natural detector does exactly this on the most capable models, while a detector built on evidence of agreement stays silent when it should. The reframing is the safety contribution: monitor for the fingerprints of a meeting of minds, not for the fact that two agents look alike.

Limitations. We are deliberately clear about what this study does and does not establish. - Sample size. Frontier runs used 2 seeds for cost reasons, so the Q4-calibrated threshold is noisy and the frontier recall figures are lower bounds, not stable estimates. The specificity result (FPR 0.0) is the robust claim; the recall figures are not. - Calibration dependence. Recall is bottlenecked by the Q4-empirical threshold. A threshold derived analytically from the economics, independent of a noisy empirical null, is the natural fix and is likely to recover recall without touching specificity. - Weaker frontier control. Hosted proxies do not guarantee a fixed checkpoint or honored seeds, so for frontier models Q3 is same model, no agreement rather than the byte-identical control we obtain by self-hosting. This weakens, but does not invalidate, the frontier negative control. - Scope. One environment (pricing) and one confounder (shared base model). Shared prompts, shared demand shocks, and other common causes are not separately tested, and transferability to negotiation or workplace settings is argued, not shown. - Interpretation if assumptions fail. Our headline claim assumes the Q3 construction is truly agreement-free; we support this with the leak-free prompt check, but if a hidden coordination affordance remained, the auditor's Q3 rejection would be understating a real signal rather than correctly rejecting a non-signal.

Future work. The clear next steps, in order of value: replace the Q4-empirical pf1 threshold with an analytic one and re-measure recall; increase frontier seeds to obtain confidence intervals; add an ablation that distinguishes pf1 from a naive high-price test by introducing a high-but-independent condition; extend the four plus factors to negotiation and workplace environments; and test multiple confounders jointly. Completing a rate-limited reasoning

model (DeepSeek-R1) would add a further frontier data point.

6. Conclusion

Collusion detectors that measure how alike two agents behave will misfire, because similarity has an innocent explanation: shared training. Antitrust law resolved this long ago by refusing to infer an agreement from parallel conduct alone and instead requiring plus factors. We turned that doctrine into a black-box auditor and tested it on six models. A standard correlation-and-judge detector false-positived on innocent parallelism, and on the two most capable models it flagged even independent agents as colluding. Our plus-factor auditor never false-positived, on any model. Its sensitivity is currently bottlenecked by a calibration choice we identify and know how to fix. The durable message is a reframing: to detect AI collusion, look for evidence of an agreement, not for the fact that two agents happen to act alike.

Code and Data

Code repository: [add link to the plus-factor-forensics repository] Data: real trajectories and per-model results for all six models are saved as JSON with accompanying figures, available via the repository.

Originality of the code. The pricing environment, the four-condition generator, both detectors, and the evaluation were written from scratch for this project. We did not clone or vendor the hackathon starter repository (github.com/poseidon-research/collusion_hackathon) or any other collusion codebase; our package depends only on standard scientific Python libraries plus the inference backends we wrote.

Tools, models, and external resources we used, with credit. Models: Qwen2.5-7B-Instruct (Alibaba/Qwen) and Llama-3.1-8B-Instruct (Meta), the latter accessed via the public NousResearch Hugging Face mirror; frontier models (GPT-5.5, Gemini-3.8-flash, Gemini-2.5-pro, Claude Sonnet 4.6) via the Kaggle Benchmarks Model Proxy. Inference: vLLM and Hugging Face Transformers on a Kaggle T4 GPU. Platform: Kaggle Notebooks and the Kaggle Benchmarks library (github.com/Kaggle/kaggle-benchmarks). Libraries: numpy, scipy, matplotlib, pydantic. The economics follow the logit-demand repeated-Bertrand setup standard in this literature (Calvano et al., 2020; Fish et al., 2024).

Author Contributions

[Fill in.]

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Appendix

A. Reproduction. The four-condition configuration, prompts, economics parameters (grid, logit demand, Nash and monopoly reference indices), and per-model figures are in the repository and the research log. Every run is seeded and reproducible.

B. Correctness checks. During development an independent methodology review verified three properties: the pf1 counterfactual replay recomputes payoff against a pinned non-cooperating rival and never reads the logged payoff; the Q3 control is leak-free (incentive and channel text appear only in Q1 and Q2, confirmed by a banned-phrase scan over all prompts); and neither detector reads the ground-truth label (verified by a label-flip test).

C. Per-model detail. pf1 firing thresholds ranged from 0.19 (GPT-5.5) to 4.22 (Llama), illustrating the calibration sensitivity discussed in Section 4.4. On Qwen the four-way ordering of mean pf1 gaps was Q2 > Q1 > Q3 > Q4, matching the intended ordering of collusion strength.